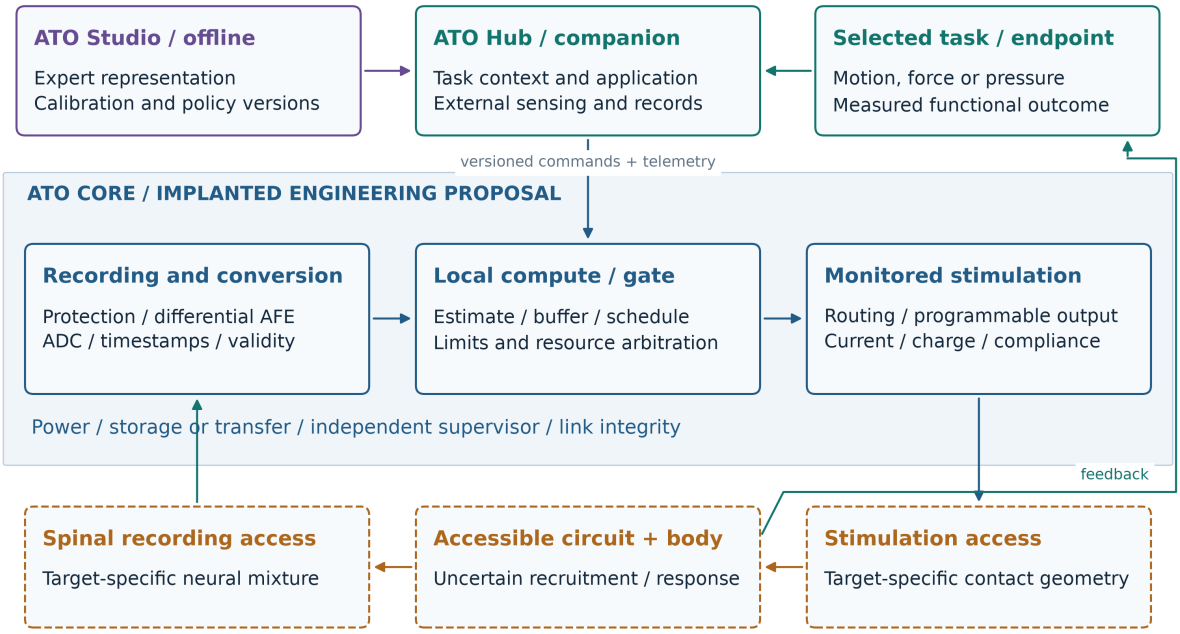
From a future spinal interface to a bounded design conclusion

Expert-informed motor practice and target-specific physiological feedback

Pratham / Project ATO / 12 October 2026

CONSERVATIVE QUALIFICATION STUDY COMPLETE



Abstract. Project ATO proposes implanted spinal recording, computational interpretation, learner-specific stimulation and independently observed feedback. Stage 3 tests the intervening digital mechanisms, information limits, numerical sensitivity and qualification rules. The historical motor pilot withholds every pulse. A separately frozen causal no-dose observer now passes a finite external-position forecast allocation. Actual delivery uncertainty and learning-response interaction remain independent limits on nonzero practice. The pressure model supports conditional pressor feedback and exposes fault/recovery tradeoffs.

Connected result. The new actual-domain replay preserves 32 one-practice trials and 64 hard-OFF assessments, with zero launched profiles. The saved-data audit checks all 96 sessions and replays all 32 controllers. Nonzero delivery is not implemented in this replay: unexpected authorization records its prefix and halts before delivery.

Finite scope. The explicit source-bound root review closes this conservative qualification study. Full native electronics, firmware, populated CAD and biological efficacy remain pending.

Scope clarification, 13 October. Future concept design continues under explicit, defensible undemonstrated assumptions. This historical negative result does not establish future SCCI impossibility or complete the conditional concept investigation.

All observations here are generated by declared artificial source/body/electrical models. No new human, animal or physical-device experiment is reported. AstroSam / Mind-Link inspires the progression from future vision to inspectable engineering; complete native electronics, firmware and populated CAD remain later work.

01 / Preserve the mechanism and its limits

The future consumer mechanism begins with a specific task or physiological endpoint. Recorded spinal information would have to be informative, causal and distinguishable from artifacts; personalized stimulation would have to act selectively within observed dose/response bounds. Feedback tests whether the intended response actually occurred.

Question	Digital evidence	Boundary retained
Expert-informed motor learning	Competent nominal synthetic teacher references; learner forecasts; independent response maps and OFF comparisons.	No expert voltage replay, human learning multiplier or retained assisted benefit is established.
Spinal information value	Capacity-matched external controls, absent/redundant fields, moving contact apertures and ambiguity witnesses.	No consistent neural advantage or connected live learner spinal increment is established.
Causal no-dose prediction	Frozen measured-data observer and fresh sensor-noise confirmation within 15 mm.	External positions, known cue, persistent prefix and synthetic structural prior; no active correction qualification.
Actual motor intervention	Whole-profile electrical adapter, finite response domains and explicit uncertainty rejection.	Actual dose domains reject; learning-enabled response interaction is unqualified.
Physiological automation	Conditional external-pressure feedback, positive pressor authority and fault fallbacks.	No generic autonomic control, spinal pressure decoder or healthy-user efficacy.

Assistance and retained learning are different endpoints

A future expert reference informs a learner-specific policy after separate competence and fidelity checks. The learner must retain a measurable benefit during truly stimulation-OFF assessment. Matched practice exposure, resources and adverse/zero/dependent plasticity controls prevent a temporary assisted movement from being reported as acquired skill.

The original 28-page paper remains a research/hypothesis map. Primary experiments support selected links, not the entire expert-recording-to-retained-benefit conjunction. Original and faster teacher references that fail remain unavailable. The expert-neural arm is an offline expert representation, not qualified live learner spinal sensing.

02 / One frozen no-dose observer

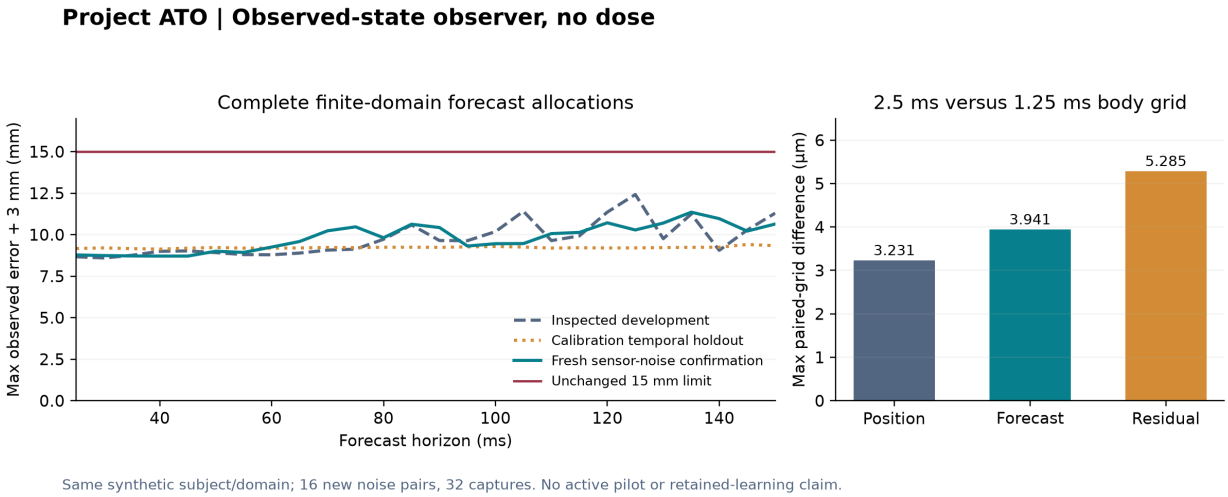


Figure 1. Saved horizon allocations from development, temporal holdout and independently predeclared fresh sensor-noise confirmation. An allocation is maximum observed coordinate residual plus the unchanged 3 mm label allowance. Covariance is not this empirical bound.

Evidence group	Complete captures passing	Largest allocation
Previously inspected drift development	16 / 16	12.418841 mm
Temporal calibration holdout	8 / 8	9.409502 mm
Fresh noise confirmation, both grids	32 / 32	11.353736 mm / 15 mm

The ten-state observer uses measured external positions, known instructions and a declared 60 ms synthetic movement prior. Calibration supplies 64 s fit exposure and 32 s temporal holdout. Private family, scale, gain, theta and future achieved motion do not select runtime parameters. State summarizes the whole accepted trial prefix, which exceeds the old stateless 75 ms history.

Confirmation retains all 550 supports and all 26 horizons for every capture: 915,200 residuals. Independence is sensor noise within the same synthetic subject and finite laws, not new-subject or biological validation. Four family names reduce to two no-dose laws. Paired 2.5/1.25 ms grids differ by at most 5.285 micrometers in residuals, with no changed budget decisions.

Resources. Ten state means, 55 unique covariance entries and 101 learned prior/drift entries; the maximum deterministic cache payload is 1,638,400 bytes, exceeding the 512 KB target RAM. This desktop result has no dose, no 25 mm combined-correction test and no MCU-ready claim.

03 / The connected replay remains explicit

The historical four-practice pilot remains unchanged: its common changing-learner forecast context fails the 15 mm gate, all 128 practice trials withhold stimulation, and all 64 OFF assessments remain pulse-free. It demonstrates no delivered assistance or retained advantage. The new observer component does not retroactively repair that result.

The new protocol has **one practice trial** per matched arm/family/grid in the actual electrical domain. This exposure is distinct from the historical four-trial protocol. A genuine pre-dose vector avoids assimilating stimulated total motion as voluntary state.

Saved evidence	Complete count / outcome
Practice / hard-OFF assessment	32 practice + 64 OFF; zero launched profiles in both.
Final frozen vector	32 sealed vectors; 832 targets, including 32 equal-time diagnostics and 800 strictly future targets.
Independent delayed guards	1600 checks: 800 warmup + 800 final future targets. Zero guard/current-diagnostic failures.
Shared baseline admission	12.418841 mm maximum / 15 mm; development + fresh confirmation. Temporal holdout grants no startup authority.
Six independently tested route patterns	192 candidate evaluations all fail actual response-domain, learning-interaction and whole-vector 25 mm gates. These are overlapping vetoes.

At 0.975 s, after supports through 0.950 s, freeze all 26 targets through 1.100 s and seal. At 1.000 s, process the delayed equal-time pre-dose diagnostic before irrevocable one-shot withholding. All updates are causal; matched arm trajectories and OFF metrics are identical.

Paired grids differ by at most 2.838302 micrometers in position, 2.458461 in frozen forecasts and 1.546171 in trajectory RMSE; no launch decision changes. Twelve distinct fresh noise arrays are matched across arms/grids. The independent audit retains all 96 sessions and 15 passing software contract probes.

OFF trajectory RMSE spans 41.925342-42.748931 mm at the same cadence and 48.360414-49.108591 mm at the faster transfer cadence. All four arms are equal within each matched case; no retained advantage is established. Kinematic peak proxies do not model force/contact mechanics.

No active delivery path is implemented here: unexpected nonzero authorization persists its decision/prefix and halts before delivery. The all-withheld replay verifies freezing, clocks, guards, withholding and hard-OFF behavior; it supplies no new nonzero delivery, post-dose monitoring/cancellation or repeated-intervention evidence. Equal OFF outcomes after no intervention do not disprove the human learning hypothesis.

04 / A prediction pass is not pulse authority

The baseline allocation stays at 15 mm, including the 3 mm label allowance; the complete correction gate stays at 25 mm. Live gain and dose intervals retain their full product, including the cross term.

Response uncertainty: $u_G|d| + |G|u_d + u_G u_d$

Add the independently observed residual at the exact physical response support and any qualified learning-interaction allowance. Expert-reference uncertainty belongs in correction against the reference; adding it to a prediction-of-actual-position residual would conceal response failure.

Independent block	Current evidence	Required consequence
Actual delivery interval	Measured fraction 1.01318359375 with uncertainty 0.4411205888605442: about 44.1% of nominal phase charge.	Every inherited actual isolated response domain rejects. Future 7.4% and 4% intervals remain assumptions.
Learning-response interaction	Inherited pulse calibration and validation used learning=False.	Nonzero learning-enabled actions require independently observed interaction coverage, or remain unavailable.
Runtime scope	No-dose qualification covers finite physical supports, cue/resources and an empty past-action prefix.	A context name cannot authorize later supports, changed action histories or arbitrary learner states.
Post-launch evidence	An actual completed monitor and delayed total-position residual arrive after the choice.	Keep commitments immutable; invalid/missing evidence latches a fault. Cancellation cannot undo a pulse.

The observer's no-dose structural prior is not a learned biological response law. After stimulation, a future multi-action system would need causal whole-prefix response correction with uncertainty or another independently qualified state model. A one-shot freeze avoids contamination for this finite experiment; it does not establish repeated active practice.

The integration is allowed to end with an unavailable active mechanism. An honest negative exit preserves the architecture and yields a concrete redesign contract without searching indefinitely for a favorable result.

05 / Physiological feedback remains central

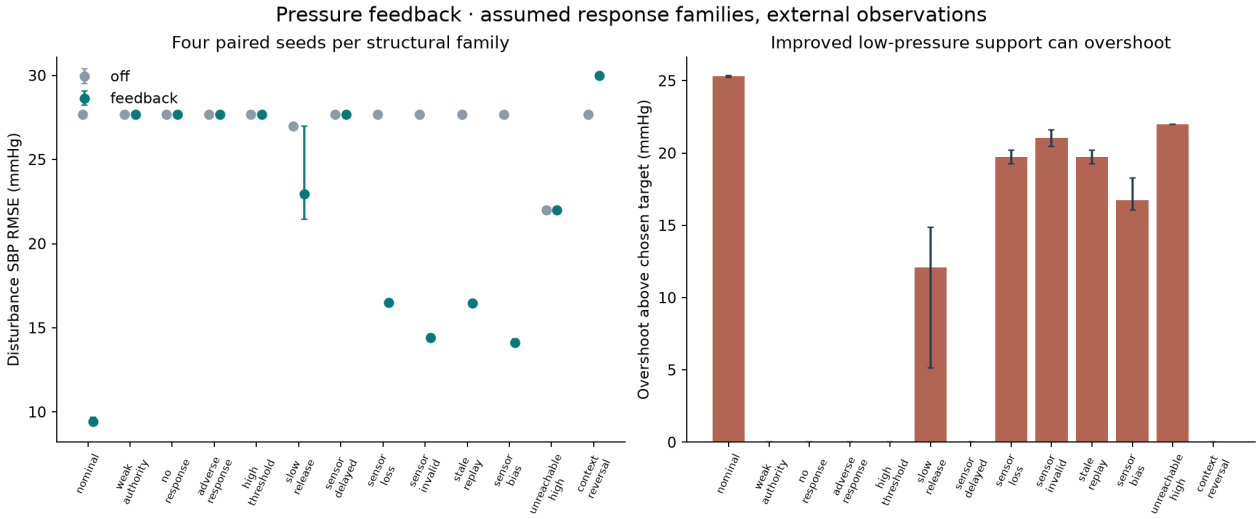


Figure 2. Preserved conditional pressure studies. Disturbance improvement can coexist with delayed recovery overshoot; fault behavior is part of the outcome. These traces use external pressure feedback and the declared reduced pressor/dose model.

A future user-specific physiological mode would select a function and accessible pathway, measure its state, identify permitted response directions, and update stimulation from observed feedback. The implemented example concerns pressure support after spinal cord injury through a positive pressor effect. It grants neither active pressure lowering nor universal control of involuntary functions.

OFF and feedback modes face disturbances, delay, altered gain, sensor loss and invalid or stale data. A privileged diagnostic oracle is excluded from benefit claims. Stop, finite-hold and ramp fallbacks expose the tradeoff between unsupported delivery and loss of support. Whole reference-train frames are admitted; fractional catch-up pulses are not invented.

Pressure refinements retain 408 runs in 136 noise-matched groups, including 24 frame-sensitive comparisons. The transduction uses an averaged monitored pulse-dose reduction and assumed recruitment. Continuous routing, analog guard behavior, real tissue response and target-specific biology remain future evidence. Dense stimulation removes qualified continuous neural recording, so the branch explicitly uses external pressure rather than inventing a neural pressure value.

06 / The electronics and CAD handoff

Responsibility	Concrete detailed-design requirement
Spinal acquisition	16 streams at 4 kHz; native differential/filter/ADC circuitry, noise/range, actual aperture/availability clocks and analog artifact recovery.
Two-source stimulation	Actual current drivers, HV polarity/contact routing, simultaneous ownership, whole-profile timing and independent shutdown.
Observed dose	Cathodic/anodic phases, net charge and recovery tails; bias/leakage, gain/offset, window/settling and interrupted-dose intervals.
Embedded control	Finite support/action domains, live uncertainty, cancellation/expiry, missing-evidence rejection, pending-response and reset/washout contracts.
Resources and package	Revised actual component power, heat, buses, memory and worst-case schedules; populate native CAD from matched boards/components/package/lead interfaces.

The first native monitoring cell has **70 components, 245 independently checked pin assignments, three editable sheets and zero ERC violations**. Its separate behavioral study contains 33 runs and 462 phase/gap/tail/net intervals, with independent KCL and matrix-exponential checks. This is a concrete partial subsystem, not the completed invasive interface.

The proposed 4% phase-charge target is unearned. The model has no shutdown actuator; bias bypass, ADC protection/settling, current-driver loops and wider waveform/load/rail behavior remain open. Two INA241 channels alone allocate 32 mW at maximum specified quiescent current, exceeding the earlier 20 mW sensing allocation. Actual system power must be revised.

The production OPA391DCKR pin mapping is corrected; its nominal SC-70 footprint still needs PCB process/land-pattern review. The restricted 24 V / 3 mA / 200 microsecond driver boundary is a reviewed proposal, with native implementation pending. A linked processor SDK example is toolchain evidence; complete ATO firmware is pending.

Stage 2 allocation solids and illustrative ports are not populated implant CAD. Stage 5 requires actual boards/components, parametric assemblies, dimensions, STEP/STL and nominal fit checks. The final integrated manuscript and reproducible private release remain later work.

07 / A finite verdict, not an efficacy claim

This conservative qualification study ends with an independently audited conclusion and a concrete hardware/software contract. A favorable learning result and resolution of all biological unknowns are not conditions to invent. A verified all-withheld replay is an acceptable negative exit; further model revisions become separately justified extensions.

CURRENT VERDICT: CONSERVATIVE QUALIFICATION STUDY COMPLETE

Supported now	Conditional or unestablished
Specified full architecture; finite synthetic acquisition/information/response studies; a frozen external no-dose forecast confirmation; conditional pressure feedback; an inspectable native monitor slice.	Useful live spinal increment, qualified nonzero learning-enabled correction, retained assistance-OFF benefit, human skill acceleration, universal body automation, full native electronics/firmware or populated implant CAD.

Future positive claims would require independent informative spinal measurements, selective and reliable stimulation, observed response/interaction coverage, matched delivered practice and pulse-free retained benefit. New evidence can narrow or overturn a specific claim without erasing the project's inspectable engineering contribution.

The explicit review is by the root AI engineering agent with a separate AI source/raw audit; it is not human, institutional or clinical approval. The separate auditor recomputes raw residuals, clocks, noise and all 192 tested candidate intervals without running the producer or controller. Its largest candidate correction allowance is 45.688604 mm against 25 mm.

Evidence and reproducibility entry points

docs/stage3/stage3-verdict.md
Full verdict and remaining decision fields

docs/stage3/observer-integration-plan.md
Reviewed one-shot clocks, scope and nonzero blocks

docs/stage3/mechanism-to-electronics-handoff.md
Concrete detailed-design requirements

experiments/stage3/observed-state-observer/verify_study.py
Frozen checks/audits; optional deterministic reproduction

experiments/stage3/frozen-observer-pilot/verify_study.py
96-session audit; 32 controller replays; same-cohort reproduction

Saved numerical input identity (SHA256 prefixes)

Observer confirmation: 35b8ce93e700afc2670a6a4f

Observer development: 02c7bcca076dd6b2739f9e31

Historical four-trial pilot: 723efd3688cc70fcbb302913

Circuit behavioral model: 836d12dfad29d33b108c2c4a

New frozen replay summary: ee4f989847e3eacb35e49143

New replay audit: 3bbc4454eac2c72b66941615

Explicit root exit review: 2cebc7e53ad2aff87a44fd12

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